PrecisionShot Training System

Anthony Fontana, DeLayne Russell, Kenn Pickavance, Nicolas Koteff

Department of Electrical and Computer Engineering
University of Central Florida, Orlando, Florida 32816

Abstract **- PrecisionShot is a standalone laser target prototype that combines optical sensing, an ESP32-S3 controller, a touchscreen, and a Bluetooth companion interface. The design seeks to convert laser pulses into scored training events while preserving local operation. Prototype work demonstrates sensor-voltage readout, local session controls, phone connectivity, and bidirectional settings synchronization. A modular enclosure aligns the sensing cells and houses the electronics. This paper summarizes the architecture, implementation, and verification approach. Complete array scanning, physical hit-to-score integration, localization accuracy, and battery endurance remain to be validated.**

Index Terms **- Bluetooth Low Energy, embedded systems, optical sensors, phototransistors, training systems, user interfaces.**

I. INTRODUCTION

PrecisionShot is a laser target training system developed by Group 13 at the University of Central Florida. Its purpose is to provide immediate feedback during dry-fire practice without making a camera or an internet connection part of the target measurement path. The original Senior Design I report establishes the motivation, subsystem tradeoffs, and engineering requirements [1]. Subsequent prototype work has concentrated on the main board, local display, phone communication, optical collection, and mechanical integration.

The central engineering problem is to turn a brief optical stimulus into one reliable training event. The target must distinguish a laser pulse from room illumination, associate the response with a physical region, update a score, and keep the local and remote interfaces consistent. These tasks are coupled: an enclosure that rejects ambient light can also reduce useful light, while extra filtering can increase acquisition delay.

The system is organized around local processing. The controller owns the session state, and the touchscreen provides access to training modes and settings. A connected phone provides remote control and another view of that state. This arrangement supports a standalone training workflow while allowing the phone interface to extend it.

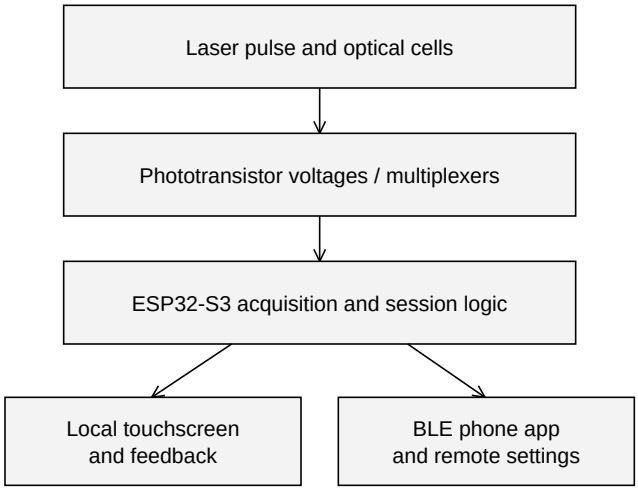


Fig. 1. Functional architecture. The optical acquisition and physical hit-to-score stages remain integration work; the display and BLE session interfaces are implemented.

II. DESIGN OBJECTIVES

The design requirements emphasize reliable detection, useful spatial accuracy, prompt feedback, and portable operation. Table 3.1 of the final report specifies that at least 90% of 30 controlled laser shots must be detected and localized within 2.0 cm. At least 95% of detected shots must appear correctly on the onboard display within 500 ms [1]. These are acceptance targets, rather than measured outcomes of the present prototype.

The mobile requirement is delivery and display within one second for at least 95% of detected shots over BLE at distances up to 10 m. The report also calls for complete installed-channel mapping, rejection of ambient-light disturbances, standalone operation, retained shot history, and four hours of stable battery operation. Section VIII describes how those requirements can be evaluated without treating simulated scores as physical laser measurements.

This paper discusses Main Board V4 firmware, the midterm prototype, and V5 R2 enclosure geometry. The original larger-array concept and the later 16-cell fit model represent different development stages. The larger array is not a tested implementation of the smaller prototype.

III. OPTICAL SENSING

A. Detection Principle

The sensing concept uses phototransistors to convert incident light into an analog voltage. Early bench work in the final report compares ambient illumination with a direct laser strike using a resistor-loaded sensor and electrical instruments. That work supports threshold detection as a starting point, but it does not establish detection probability throughout an assembled target [1]. Optical collection and electrical detection must therefore be tested together.



Fig. 2. Earlier printed four-cell enclosure prototype shown during midterm preparation. The photograph documents a physical optical-cell prototype, rather than the later V5 R2 assembly.

The original report identifies the through-hole ALS-PT204-6C/L177 as a bench component and the ALS-PT19-315C/L177/TR8 as a surface-mount selection. Preserving this distinction matters because the sensor body, active region, and aperture alignment affect how much light reaches the device. A simulation containing the larger through-hole package cannot by itself validate the response of a different package on a PCB.

B. Optical Cells and Ambient Light

The target face uses separated optical cells to associate incoming light with a sensing region. Angled walls are intended to direct useful light toward the detector while limiting off-axis illumination. The midterm work examined a two-inch chamber and found that laser strength and strike position affected detection [4]. The resulting observation is qualitative: a sensor response was visible, but uniform collection over the cell opening was not established.

A diffuser and reflective surfaces change both the useful pulse and the background received by the detector. Spreading a small laser spot can improve coverage, but also reduces the concentration at any one location. Reflection can redirect light that misses the detector; the actual benefit depends on wall finish, incidence angle, aperture obstruction, and alignment. These effects must be measured after assembly.

C. Simulation and Physical Evidence

The optical study prepared for the midterm illustrates a 50.8 mm chamber with a bottom-mounted ALS-PT204 body, a frosted entrance, and reflective-wall assumptions [5]. It compares central and displaced illumination to examine possible light paths. Its ray images are useful for explaining geometry, but their intercepted optical power is a model output. It is not an electrical sensitivity measurement or a measured probability of detecting a training pulse.

For this reason, the paper does not use simulated power to claim compliance with the accuracy or ambient-light requirements. A useful follow-up is to move a repeatable laser stimulus over marked cell locations while recording the actual sensor waveform. Center, edge, and corner positions should be included, with the same distance, pulse source, and room-light condition identified for each trial.

D. Spatial Resolution

A cell response establishes a region before it establishes a continuous coordinate. For example, a square cell at 50 mm pitch has a center-to-corner distance of approximately 35.4 mm. Reporting only its center cannot guarantee a 20 mm localization error across the whole cell. The 2.0 cm requirement therefore needs demonstrated interpolation, a smaller effective sensing region, or an explicitly characterized coverage limit.

The final report proposes retaining relative analog amplitudes for position estimation. A baseline-corrected weighted centroid is one possible method when neighboring responses contain enough information. If only one isolated channel responds, the estimator has no evidence for a subcell position. An honest localization test must compare reported coordinates with marked ground truth and include all misses in its detection accounting.

IV. ELECTRONIC ARCHITECTURE

A. Main Controller and Interfaces

Main Board V4 uses an ESP32-S3-WROOM-1-N8R8 and a four-inch ST7796S display with an FT6336 touch controller. The implemented display is operated in 480 by 320 landscape orientation. The firmware uses native ESP-IDF C++ and separates hardware access, drawing, session rules, sensor diagnostics, and Bluetooth communication [2]. This division makes it possible to exercise the display and communication paths before the optical acquisition path is complete.

TABLE I
IMPLEMENTED MAIN BOARD INTERFACES [2]

Function	Interface and connection
Display	SPI; GPIO11-17 for data, clock, selects, reset, and backlight
Touch	I2C on GPIO39/40; interrupt GPIO41 and reset GPIO42
Sensor diagnostic	ADC input on GPIO1; J_PSB pin 9
Regulator telemetry	ADC inputs on GPIO9 and GPIO10
Audio	GPIO44 drives the LM386 input
Native USB	GPIO19 and GPIO20
SD interface	Chip select GPIO18 is held inactive

The display and amplifier use the 5 V supply, while the controller uses 3.3 V. The sensor diagnostic input is a low-voltage ADC input and must be kept within the documented 0 to 3.3 V range. GPIO identifiers must be distinguished from module pad numbers; the recorded Main Board V4 bring-up corrected an earlier development-board pin map that had produced a white display [3].

B. Multiplexed Sensor Readout

The larger array concept uses CD74HC4067 multiplexers to route multiple sensor outputs to a limited number of ADC inputs [1]. This reduces direct controller pin requirements, but introduces a timed acquisition sequence. Firmware must select a channel, allow the analog path to settle, read the voltage, and retain a stable mapping from the channel address to its physical location.

For N sequentially read channels, a first-order scan budget is

$$T_{scan} = N\,(t_{select} + t_{settle} + t_{ADC}). \quad (1)$$

Here, channel selection includes control-line updates, settling includes the analog path and ADC acquisition behavior, and conversion includes the chosen read procedure. Equation (1) is a planning relation, not a measured scan time. Extra averaging, processing, and competing display or radio work must also be included when determining whether a short pulse can be missed between visits to a channel.

C. Power and Audio Integration

Regulated rails support the digital controller, optical circuitry, display, and audio loads. MCP9700-based regulator temperature readings are exposed through the diagnostic interface. Those readings help observe operating conditions, but they do not replace rail-voltage measurements, battery endurance tests, or a reference-temperature check [2], [3].

The two piezo speakers share the LM386 output. Debug animations can deliberately play short tunes, with nonblocking sequencing and cancellation in the firmware. An unresolved prototype observation is audible noise associated with display updates. Software changes that stop idle PWM did not resolve that reported symptom; amplifier gain and power or ground coupling require physical investigation [3].

D. Hardware Revision Boundaries

The project archive contains tiled arrays, higher-density board studies, and a later sensor-only schematic. These are design alternatives and revision evidence, rather than interchangeable descriptions of the demonstrated unit. The V5 R2 enclosure is fitted to supplied standalone electronics models: a 200 by 255 mm sensor PCB and a 210 by 110 mm main PCB [6]. Its 16-cell arrangement should not be confused with the original 500-position concept or the separate 256-detector study.

Electrical and mechanical interfaces must be frozen together before final testing. In particular, a valid channel map depends on the actual populated board, and any change in sensor pitch changes the mapping used for localization. The same revision identifiers should appear in the assembly record, firmware pin map, optical calibration record, and test data.

V. EMBEDDED SOFTWARE

A. Session State and Local Control

The firmware separates scoring rules from screen rendering. Freestyle continues an open session, whereas Classic collects ten shots and then displays a total out of 100. The completed Classic total remains visible until the next shot starts a new round. Reset clears the session, and a change of mode starts a fresh session [2]. These rules can be checked using injected test scores without requiring the optical hardware.

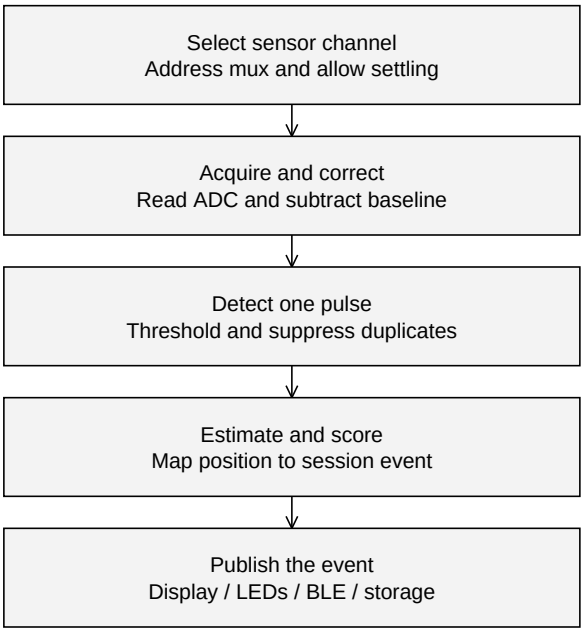


Fig. 3. Intended physical-shot processing sequence. The present diagnostic ADC path does not execute this complete array-scanning and scoring pipeline.

Local controls provide mode, distance, units, sensitivity, theme, Bluetooth, and diagnostics. Distance is stored in millimeters, preserving physical distance when units change. Sensitivity is bounded to the raw ADC range. These settings support operation, but do not establish a completed per-channel optical calibration.

B. Diagnostic Readout

The sensor module presently reads three ADC channels: two regulator-temperature outputs and the shared sensor diagnostic input. Each update averages 16 conversions per channel, with updates scheduled every 200 ms. Calibrated millivolts are reported when ADC calibration is available; raw values remain available if calibration cannot be created [2]. This is a diagnostic measurement path, rather than a complete high-speed scan of the target array.

The debug detection indicator compares the sensor input with the configured raw threshold. It does not add a scored shot. Keeping those paths separate avoids mistaking a visible threshold crossing for a fully localized training event. It also allows electrical bring-up to proceed while the optical geometry and scanning method are still being refined.

C. Physical Hit Integration

A complete acquisition path must convert a pulse into exactly one event. The proposed sequence in Fig. 3 includes baseline subtraction, pulse detection, position estimation, scoring, and publication. A shared event record should carry the time, sensor evidence, computed position, score, and active mode. Local display and BLE publication should consume the same record so that they cannot independently assign different scores to one pulse.

Baseline calibration should sample every installed channel under the chosen ambient condition. It should detect invalid, saturated, or disconnected channels rather than silently treating them as ordinary low signals. The detector then needs a documented threshold margin and a recovery or duplicate-suppression rule. A long pulse spanning several scans must not be counted repeatedly, while a short inter-shot interval must not be merged into one event.

Timing must be measured at the acquisition boundary. The current 200 ms diagnostic interval is useful for observing voltage levels, but a brief training pulse can occur between samples. A faster scanning implementation needs its own settling and pulse-width tests before the existing session UI can be described as a live laser scoring system.

D. Scheduling and Persistence

The interface, Bluetooth queue, sensor sampling, and sound routines share the embedded runtime. Nonblocking sound and animation timing preserve the ability to handle other work, but they do not prove that array acquisition deadlines are met. The integration test should include active display changes and communication traffic while observing scan timing.

Session state and settings currently reset after restart. SD logging, battery readings, and persistent history remain planned [2]. Persistence needs explicit recovery behavior for interrupted writes and for distinguishing a new session from a resumed one.

VI. BLUETOOTH AND PHONE INTERFACE

A. Shared State Model

The phone connects to a device advertised as PrecisionShot. A TX characteristic provides reads and notifications; RX accepts commands. Local actions and phone requests share the embedded state so mode and theme changes propagate between interfaces [7].



Fig. 4. Capture from the midterm video review showing the phone interface, Main Board V4 touchscreen, and power setup. The shown session flow used debug scores; this image does not demonstrate physical laser localization.

The original compact shot message remains a JSON record containing a hit number and score. It does not contain physical coordinates. Protocol version 2 adds full state snapshots for mode, count, total, distance, sensitivity, page, theme, and diagnostics. Each notification record is kept to at most 20 bytes, and a snapshot is bounded by matching BEGIN and END sequence values [7].

B. Commands and Confirmation

The app can send a numbered command such as @12:MODE:CLASSIC. The board responds with state and ACK:12 when the command has been accepted and queued, or a numbered error when it cannot be accepted. The app waits for a complete snapshot before presenting the new state. This avoids rendering a mixture of old and new fields while a sequence is still arriving.

Invalid values are rejected without corrupting the previous setting. A non-idempotent test-shot command should not be automatically repeated after a timeout because it might already have changed the session. The protocol instructions instead call for requesting state first. A command acknowledgement establishes application-level handling; it should not be treated as confirmation of an optical measurement.

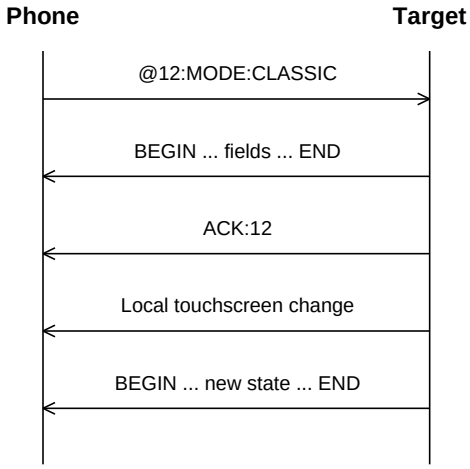


Fig. 5. Command confirmation and shared-state publication. The same state publication path also reports local touchscreen changes.

C. Diagnostics and Reconnection

The phone requests sensor telemetry through WATCH:1. WATCH:0, disconnection, or lease expiry stops that traffic. The ten-second lease prevents an abandoned debug view from leaving a continuing sensor stream. Core session state remains available independently of this optional stream. State requests do not force an unnecessary display redraw [3], [7].

Recorded acceptance work includes reconnecting, resubscribing, malformed-command rejection, mode changes, score rollover, settings synchronization, and command confirmations. The October connected-board run passed 71 confirmed commands with 1,517 notifications. A later watch/synchronization check recorded 30 matched confirmations and a maximum observed confirmation latency of 0.391 s [3].

Those results establish communication behavior under the documented test conditions. They do not establish the one-second physical-shot delivery requirement at 10 m, because they did not measure a controlled laser event through sensing, scoring, transmission, and app rendering. That complete path remains part of the acceptance procedure.

Retained session history requires a separate persistence test. A live score received correctly is not evidence that it survives closing and reopening the app.

VII. MECHANICAL DESIGN

A. Enclosure and Electronics Alignment

The enclosure performs an optical role as well as a protective one. It fixes each cell relative to its detector, keeps the display accessible, and houses the electronics. Figure 6 shows Nick's original V4 enclosure parts. The subsequent V5 R2 mounting study discussed below uses the supplied standalone electronics models as its fit authority [6].

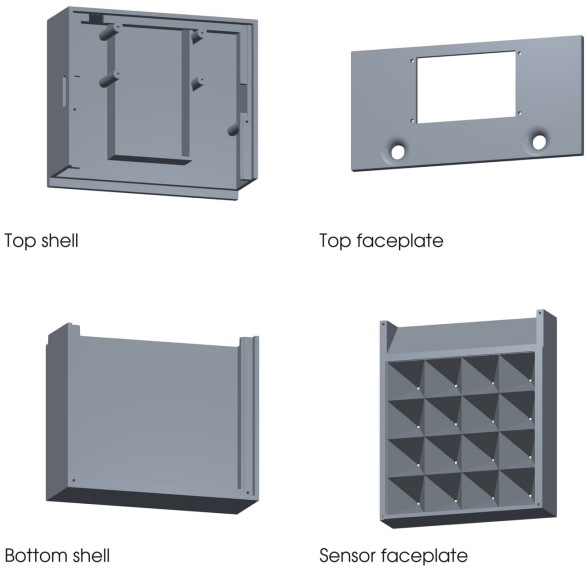


Fig. 6. Nick's V4 enclosure: top and bottom shells (left), display and sensor faceplates (right). Rendered from the original STEP files supplied Oct. 26, 2026; views show separate parts, not a tested assembly.

The sensor-board fit model has sixteen sensing cells in a four-by-four arrangement at 50 mm pitch. Two eight-cell cassettes preserve the angled chamber walls. The mouth-to-detector-plane distance is 50.8 mm. Small detector openings reduce exposed PCB area, while plain wall cutouts expose the adjacent LEDs [6]. Their effect on optical collection requires measurement rather than inference from the model.

B. Mounting and Service Access

The hole-free sensor PCB is captured along its perimeter with slots, removable edge caps, and end stops. This avoids introducing new mounting holes into the reference board. The display remains connected through its header, while four shared screen and main-board screws clamp posts and spacers. The mounting stack is intended to carry enclosure loads through the fasteners rather than through the electrical header.

The design uses a confirmed 10.2 mm separation from the main PCB front surface to the underside of the display PCB, including the seating gap. The actual screen stack should be checked before tightening because connector tolerances and printed spacer fit are not established by a collision-free CAD model [6].

Two regulator modules occupy an unused strip near the sensor board. The battery sits behind the main-board region in a rear-loaded cradle. Removable bridges retain the battery case while leaving a window for connector access. The model uses a seller-listed battery envelope; exact port placement and actual case dimensions remain physical fit checks.

C. Fabrication and Assembly

The overall V5 R2 model is approximately 242.8 mm wide, 422 mm high, and 92.4 mm deep, including external seam straps. It is split into parts that fit a 256 mm printer bed. Nominal shell walls are 1.0 mm, with local bosses at fastening points. Side-loaded captive M3 nuts keep front mounting hardware behind a continuous skin [6].

A separate fit coupon checks the sensor-board slot, screws, and nuts before printing the full set. The saved printer projects define part orientation and arrangement. Slicer estimates and digital collision checks support fabrication planning, but they do not establish printed strength, fastening life, or clearance after thermal distortion.

The ribbon connection has a reserved corridor between the board regions. Cable routing must preserve access to plugs and bends without obstructing the battery, vents, or optical cells. A bench check should precede enclosed operation so that wiring errors, amplifier noise, or regulator heating are observable while the electronics remain accessible.

D. Revision Validation

The earlier printed cells, Nick's V4 CAD parts, and the V5 R2 mounting study document distinct development stages. The photograph establishes fabrication of an optical-cell prototype; the CAD views show design geometry. Final reporting should tie each physical test to its exact enclosure and board revision, without treating a render as evidence of assembly or calibration.

VIII. VERIFICATION AND OBSERVATIONS

A. Prototype Evidence

The midterm observations demonstrate sensor readings under flashlight and direct-laser illumination, phone discovery and connection, settings synchronization in both directions, and score changes produced by simulated hits [4]. The two-inch diffuser arrangement still depended on strike position and laser strength. These observations identify working interfaces and an optical tuning problem, rather than a completed detection-performance test.

TABLE II
RECORDED EVIDENCE AND ITS LIMIT [3], [4]

Evidence	Supported conclusion
Sensor-voltage response	A light stimulus can change the diagnostic reading; full coverage is unmeasured.
Classic / Freestyle debug tests	Session rules respond to injected scores; no physical coordinate is established.
71 confirmed commands; 1,517 notifications	Connected command/state handling passed the recorded run.
0.391 s maximum in a 30-confirmation watch test	Observed command confirmation time; not laser-to-app latency.
45-48 ms frame render and transfer in earlier UI checks	A display subsystem measurement; not complete hit-to-display timing.
Printed cells and CAD checks	Prototype fabrication and digital geometry evidence; not enclosure acceptance.

B. Detection and Localization

The principal acceptance series uses 30 controlled laser shots at predefined coordinates [1]. A fixed reference grid should cover the center, edges, and corners of the active area. Each attempt needs an identifier, ground-truth position, lighting condition, distance, detection result, and reported position. The required combined detection/localization result is at least 27 successful attempts out of 30 within 2.0 cm.

A miss must remain in the denominator. Reporting localization only for detected shots can hide weak coverage. Duplicate detections should also be counted separately, since one pulse producing two scores violates the intended event model even if one reported coordinate is correct. The test record should retain raw channel evidence where possible so a failed trial can be traced to optical collection, thresholding, mapping, or estimation.

C. Timing and Data Integrity

The display test must start from a documented detection timestamp and end when the correct shot information is visible. The BLE test extends the same event to the phone display. With 30 detected events, a 95% target requires at least 29 timely events. A larger series would provide a stronger view of tail delays, but the recorded pass criterion should not change after results are observed.

Screen render duration and command acknowledgement duration are useful subsystem timings. Neither includes the entire path from an incident laser pulse. A trace that marks detection, scoring, display completion, transmission, and app rendering will expose which stage dominates delay. The clock alignment or instrument method must be recorded so that timestamps from different devices can be compared.

D. Remaining Acceptance Series

Ambient-light testing repeats the shot series under dark, normal, and bright indoor conditions, followed by a ten-minute idle interval in each condition. The requirement is at least 90% detection reliability and no more than one false shot during the idle test [1]. Room-light descriptions should be supplemented by repeatable setup information, because the labels alone do not define a reproducible optical condition.

Channel mapping requires every installed sensor to be readable at the correct physical location. Standalone testing disconnects the phone and exercises an entire local session. Persistence testing closes and reopens the app after at least 100 records. Battery testing runs the complete system for four continuous hours while monitoring resets and both supply rails. The 5 V and 3.3 V rails must remain within 5% of nominal [1].

No completed series in the available evidence establishes those full-system requirements. The prototype results therefore support a working control and diagnostic platform, with physical acquisition, optical uniformity, persistence, and endurance still requiring measured acceptance tests.

IX. DISCUSSION

The prototype has reduced uncertainty in the user interface and communication path. Local session rules can be exercised independently of the detector, and the phone can configure and observe the same embedded state. That separation makes failures easier to isolate: an incorrect Classic total can be investigated in session logic, while a missed physical pulse can be investigated in optical acquisition without rebuilding the app.

The largest remaining uncertainty lies between optical collection and a valid scored event. A sensor-voltage response is a necessary step, but the useful target must perform across an area, under changing illumination, and over many pulses. The physical test must also establish that different cell locations produce enough distinguishable evidence to meet the localization requirement. Increasing channel count alone does not prove accuracy.

The enclosure and firmware should be evaluated as one measurement system. Narrow apertures may reject background light while losing edge strikes. Reflective surfaces may improve collection while increasing cross-cell response. Longer averaging may stabilize a diagnostic number while missing a brief pulse. These tradeoffs should be resolved using waveform and position data from the assembled revision.

The current evidence also shows the importance of separating test scope. A successful BLE command test exercises request handling and state delivery. A display transfer time measures screen work. A CAD intersection audit measures geometry. Each is useful, but none can substitute for the full optical-to-score experiment or a sustained battery run.

A. Integration Priorities

The next integration step is a repeatable single-cell pulse measurement, followed by a documented scan over every installed channel. That sequence should establish signal polarity, ambient baseline, peak amplitude, settling time, pulse width, and duplicate suppression before connecting the acquisition result to session scoring. The coordinate mapping and calibration record can then be frozen for the first controlled 30-shot series.

After physical scoring is connected, the same event should be traced through the local display and phone. Persistence and battery tests can then use real sessions with the intended display, radio, and optical workloads. This sequence prevents an otherwise convincing interface demonstration from obscuring unfinished measurement behavior.

X. CONCLUSION

PrecisionShot combines an optical target concept with an implemented ESP32-S3 control platform, a local touchscreen, and a bidirectional Bluetooth phone interface. Prototype evidence supports sensor diagnostics, session scoring with injected hits, local and remote settings, and physical optical-cell exploration. The V5 R2 enclosure provides a concrete geometry for integrating the electronics, optical chambers, display, and battery.

The remaining work is to establish reliable physical acquisition and carry each detected pulse through localization, scoring, feedback, and storage. The specified accuracy, ambient-light rejection, response time, wireless range, persistence, and battery endurance still require complete measured series. Meeting those criteria will determine whether the current prototype becomes a repeatable standalone training target.

REFERENCES

[1] A. Fontana, D. Russell, K. Pickavance, and N. Koteff, *PrecisionShot Training System Senior Design I Final Report*, Group 13, University of Central Florida, Summer 2026, chs. 2-7 and Table 3.1.

[2] PrecisionShot team, *PrecisionShot firmware*, README and C++ sources, including sensors.cpp and session.cpp, local project snapshot reviewed Oct. 26, 2026.

[3] PrecisionShot team, *Training UI validation*, firmware validation record, entries Sept. 11-Oct. 5, 2026.

[4] PrecisionShot team, *Midterm summary*, test observations and prototype status slides, Oct. 17, 2026.

[5] PrecisionShot team, *Reflection snapshots current model*, optical simulation assumptions and capture manifest, Oct. 17, 2026.

[6] PrecisionShot team, *PrecisionShot enclosure V5 R2 slim mounting prototype*, README, parameters, source geometry, and CAD previews, Oct. 26, 2026.

[7] PrecisionShot team, *Bluetooth commands*, protocol version 2 and WATCH telemetry documentation, local firmware snapshot reviewed Oct. 26, 2026.